

Chapter 2

Atoms and the Stern-Gerlach Experiment

Learning goals:

- (1) Explain how the models for the atom evolved as additional experimental information was discovered
- (2) Calculate the effect of uniform and inhomogeneous magnetic fields on an atomic beam
- (3) Describe the reasons why the experimental result for the Stern-Gerlach experiment cannot be explained classically
- (4) Use the concept of indeterminacy (superposition) to analyze the results of multiple analyzer-loop experiments
- (5) Explain how the concept of entanglement is used for a Stern-Gerlach experiment

Material to review:

- (a) Uniform circular motion
- (b) The Lorentz force law
- (c) The behavior of current loops in a magnetic field
- (d) Precession
- (e) Angular momentum
- (f) How accelerating charges radiate energy

2.1 Early atomic models

In Chapter 1, we showed how experimental evidence established the existence of the photon and of the electron. At the dawn of quantum mechanics, it was also known that atoms existed, and their approximate size and mass, but much about their structure was unknown. In particular, the proton had not yet been discovered. So, because the electron mass was thousands of times less than the mass of the atom, one

could imagine many electrons in the atoms, even for the simplest element hydrogen. Of course, atoms are neutral, so there must be some form of positive charge in the atom as well.

John J. Thomson used these ideas to construct what is called the plum-pudding model of the atom [11]. It assumed that there was some uniform background positive charge, and the electrons were bound inside this positive charge in a stable static configuration. One could imagine them held within harmonic traps as we discussed previously. The main benefit of this model is that it is quite stable. There were other ideas floating around as well, but we will not discuss those in any great detail.

Ernest Rutherford (see Fig. 2.1) set out to see what was inside the atom [8]. He used a radioactive element that emitted alpha particles, which are helium nuclei. These are positively charged particles that have a high velocity. In the plum pudding model, they should be deflected slightly, because they would feel the inhomogeneity of the charge distribution inside the atom, and this inhomogeneity was not expected to be large, because the charges are so spread out. But, this is not what was seen. Instead, it was found that the positive charge was concentrated in a nucleus, whose size was 100,000 times smaller than the atom itself. One key observation supporting this picture was that the alpha particles backscattered from the gold film target. This means that they were reflected back from the foil. Rutherford likened this to shooting an artillery shell onto a tissue paper and seeing it bounce back. This is because the target he used was thin gold leaf, which was flattened to a very narrow width, and the alpha particles moved at a very high velocity, approaching a fraction of the speed of light. So, backscattering seemed to be highly unlikely. We analyze this scattering problem next. It turns out that the classical treatment exactly agrees with the quantum treatment, which is an amazing coincidence. We will see some more amazing coincidences like this for other problems as well. This allowed the classical analysis to properly conclude results about the nucleus that were correct even with a much later full quantum treatment.

We start by setting up how a scattering experiment works because this is something you are probably not familiar with. The basic idea for the experiment is simple—shine a beam of particles onto the target and measure what comes out. So, we have an initial particle, with velocity $\vec{v}$ moving toward the target. We consider the case of the target being a point object and our beam particle being a perpendicular distance b away from the center (called the impact parameter). By varying over a range of b values, we will cover all the particles in the beam. After scattering, the beam particle will be measured emerging at some angle θ relative to the incident direction determined by $\vec{v}$. The goal of a theory for the scattering process is to use the force law between the beam particle and the target to determine the angle of the scattering. If this sounds difficult, it is. But, we really can solve it as we are about show you. A schematic figure showing this geometry is plotted in Fig. 2.2

Analyzing the Rutherford scattering problem is challenging to complete. We will go through it in detail here. You know all of the pieces you need to know to work this out, but there are many steps. You might not be used to solving problems like this, so you may get discouraged before you are finished. Please stick with it, as you can make it through. In fact, the process is similar to cooking in the traditional French

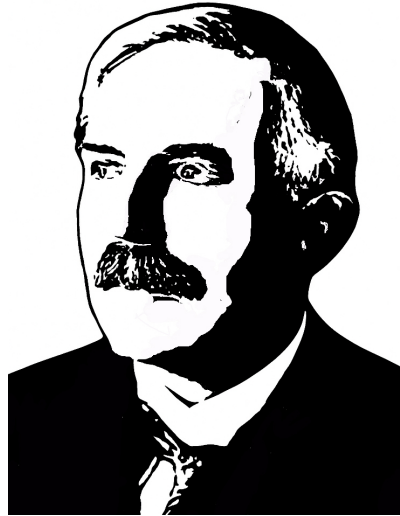


Fig. 2.1: Ernest Rutherford in the 1920's. Rutherford worked with Geiger and Marsden to perform the scattering experiment, which led to the discovery of the nucleus of the atom. He won the Nobel prize in 1908. Later, Niels Bohr analyzed the data to provide the first quantum theory of the atom. [Library of Congress, George Bain Collection, LC-B2- 6101-9. Public domain. Converted to look like a stencil by the authors.]

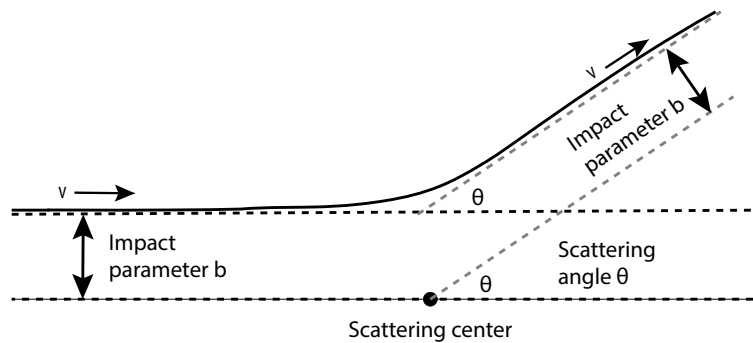


Fig. 2.2: Schematic of the scattering problem. A particle with an initial horizontal velocity $\vec{v}$, which is a perpendicular distance of b away from the scattering center, will be scattered into a velocity moving in the direction θ away from the scattering center when measured far away in a detector.

style. As an example, we decided to cook a Paris-Brest cake (named after a famous bike race in France). We followed the Jacques Pepin recipe. To make the cake one first makes an almond brittle, then you use a mortar and pestle to grind it into a praline. Next, you make a large roux which is shaped into a bicycle wheel shape, covered with almond slices, baked, and set aside. Next, you make a vanilla custard (with vanilla beans, of course). Finally you make a whipped cream (whipped by hand). Then everything is assembled—cut the cake in half, put down a custard layer, top with praline, then whipped cream, and put the top on. Dust with powdered sugar and it is ready to go. It took nearly two days to make. No step was hard, there just were many, many steps. We call this “French cooking.” Many problems you need to solve in physics are French cooking problems. No steps are too hard, there just are many of them. Rutherford scattering is a French-cooking problem. But, it is always worth it to get to the end—indeed, the cake was wonderful!

To start solving this problem, we need to focus on two conserved quantities. The first is the energy, which is entirely kinetic, when the beam particle is far from the scattering center, so it satisfies $E = \frac{1}{2}mv^2$ assuming the mass of the beam particle is m . The second is the angular momentum about the scattering center, if the force from the scattering center is a central force. The initial angular momentum is $L = mvb$, because the impact parameter b is the perpendicular distance of the beam particle from the scattering center. For Rutherford scattering, we are working with the Coulomb problem. The alpha particle has a charge of $2e$, while the scattering center has a charge of Ze . The force between the two particles is $Ze^2/2\pi\epsilon_0 r^2$ with r the distance between the two particles, and ϵ_0 the vacuum permittivity. If the impact parameter is zero, then the particle will move to a distance of closest approach given by the moment when the initial kinetic energy is converted entirely to potential energy of the Coulomb interaction (note we assume the scattering center is fixed in space and does not move). You should be able to show that the distance of closest approach is given by $D = Ze^2/\pi\epsilon_0 mv^2$. Note how higher initial energy is needed to investigate smaller and smaller distances, which is why particle accelerators always strive for higher and higher energy.

Our first job is to *estimate* the relationship between b and θ . Estimates are a good first step to make in any problem, as they help you know if your solution is on the right track. Estimating can be done in the case where θ is small, similar to what we did in Chap. 1 when we estimated the deflection in the m/e experiment. We approximate the force as acting over a short period of time determined by the impact parameter b . So we assume we have a constant force of $Ze^2/2\pi\epsilon_0 b^2$ acting over the time $t = 2b/v$. Then the perpendicular momentum given to the particle is the force times the time or $p_\perp = Ze^2/\pi\epsilon_0 bv$. Using the distance of closest approach, we can rewrite this as $p_\perp = 2DE/bv$. If b is large, so the angle is small, then we have

$$\theta \approx \frac{p_\perp}{p} = \frac{D}{b}, \quad (2.1)$$

or $b = D/\theta$. We will see the exact formula reduces to this in the large b limit. Let us go ahead and find that exact result. Note that our estimate is an order of magnitude

estimate, so we do not expect exact agreement, but only agreement up to a constant factor. We will see how well we did later.

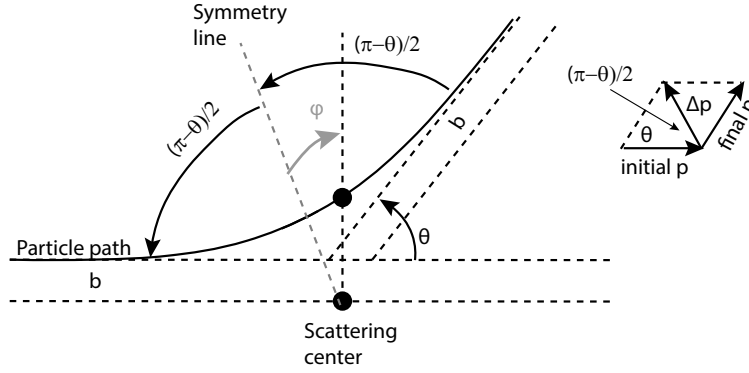


Fig. 2.3: Geometry for the Rutherford scattering. The path of the particle is the curved solid black line. The scattering angle is θ . The gray dashed line is the line of symmetry, which bisects the scattering path and points in the direction of the change in momentum. We draw a line from the instantaneous position of the particle (here vertical to the scattering center) to the scattering center and define the angle ϕ to be the angle between the drawn line and the line of symmetry. On the right, we show the parallelogram with the initial, final and change of momentum.

We need to carefully construct a figure that summarizes the scattering process. The geometry is given in Fig. 2.3. The scattering angle is θ . The line that bisects the remaining angle $\pi - \theta$ is called the symmetry line. We use the angle ϕ from the symmetry line to a line that goes through the position of the particle and the scattering center as our coordinate for the particle as it moves along its scattering path. The initial momentum is horizontal, the final momentum is in the θ direction, and the change in momentum lies along the symmetry line. The initial and final momenta have the same magnitude, but the transferred momentum can have a different magnitude, so the momentum triangle formed by $\vec{p}_i$, $\vec{p}_f$ and $\Delta\vec{p}$ is an isosceles triangle, with one angle θ and two angles $(\pi - \theta)/2$. Dropping the perpendicular that bisects the θ angle shows us that

$$\frac{\Delta p}{2} = p \sin \frac{\theta}{2}, \quad (2.2)$$

where $p = |\vec{p}_i| = |\vec{p}_f|$ is the magnitude of the initial and final momentum. Our job is to find the magnitude of the total momentum that is transferred Δp , which we find by integrating the component of the force along the direction given by the symmetry line over time. This gives us

$$\Delta p = \int_0^\infty dt \frac{Ze^2}{2\pi\epsilon_0 r^2(t)} \cos \phi(t) = \int_{-\infty}^\infty dt \frac{DE}{r^2(t)} \cos \phi(t), \quad (2.3)$$

where $r(t)$ is the distance of the particle from the scattering center, and $\phi(t)$ is the angle to the symmetry line. We use angular momentum to simplify the expression. We have $mr^2(t)\dot{\phi}(t) = L = pb$ because angular momentum is conserved. We change the integration from time to ϕ using $dt = d\phi/\dot{\phi}$ with angular momentum conservation expressed as $r^2\dot{\phi} = pb/m$ to find that

$$\Delta p = \int_{\phi_i}^{\phi_f} d\phi \frac{DEm}{pb} \cos \phi = \frac{Dp}{2b} (\sin \phi_i - \sin \phi_f). \quad (2.4)$$

Using the fact that $\phi_i = -(\pi - \theta)/2$ and $\phi_f = (\pi - \theta)/2$ then tells us that we have

$$\Delta p = \frac{Dp}{b} \cos \frac{\theta}{2}. \quad (2.5)$$

Using the fact that $\Delta p = 2p \sin \frac{\theta}{2}$ then gives our final result:

$$b = \frac{D}{2 \tan \frac{\theta}{2}}. \quad (2.6)$$

This is a remarkably simple result for what looked like an intractable problem at first! It also agrees exactly with our estimate if we work with small θ and use the fact that $2 \tan \frac{\theta}{2} \approx \theta$.

Relating the impact parameter to the scattering angle is not quite enough to completely analyze the scattering experiment. This is because we will send in a beam that has a certain number of particles per unit area per unit time traveling in it (this is called a flux F of particles) and we want to count the number that can be found scattering at an angle of θ in our detector, looking at the fraction that scatters as a function of θ . This full analysis allows us to properly interpret the data. So, we need to work a bit harder to determine these fractions.

Assume that our beam has a circular area. Then, the differential number of particles in the beam between an impact parameter of b and $b + db$ is given by $N(b) = F2\pi b db$; this comes from differentiating $F\pi b^2$, or can be inferred by looking at the difference of areas between a circle of radius b and $b + db$. We want to convert this from an expression in terms of the impact parameter, which is hard to measure, into an expression in terms of the scattering angle, which is easy to measure—this means we want to measure the particles scattered between θ and $\theta + d\theta$. As we mentioned before, what we really want is the fraction of particles scattered. This is called the differential scattering cross section and is given by

$$\frac{d\sigma}{d\theta} = \frac{1}{F} \frac{dN(\theta)}{d\theta}. \quad (2.7)$$

Computing this just requires the chain rule and using our relation in Eq. (2.6). We find

$$\frac{d\sigma}{d\theta} = 2\pi b \frac{db}{d\theta} = 2\pi \frac{D}{2 \tan \frac{\theta}{2}} \frac{d}{d\theta} \frac{D}{2 \tan \frac{\theta}{2}} = \frac{\pi D^2}{4} \frac{\cos \frac{\theta}{2}}{\sin^3 \frac{\theta}{2}}. \quad (2.8)$$

Be sure to verify this result yourself. We next have to grapple with the fact that the scattering is in three dimensions and we have considered the scattering in a two-dimensional plane only. Converting to three dimensions requires us to go over to the solid angle, rather than the scattering angle. Geometrically, this corresponds to considering the cone of scattering that is at an angle of θ to the beam axis and rotated around the axis by an angle of 2π . The solid angle is then $d\Omega = 2\pi d \cos \theta = 2\pi \sin \theta d\theta$, as you should know from integration in spherical coordinates. This then yields our final result

$$\frac{d\sigma}{d\Omega} = \frac{D^2}{16 \sin^4 \frac{\theta}{2}}. \quad (2.9)$$

Note how the result is finite for backscattering, which occurs at $\theta = \pi$. Curiously, the scattering becomes infinite for $\theta = 0$, which corresponds to the large b range. But, note that for scattering off of a solid, the value of b can never be larger than half the distance between the atoms in the solid, so b cannot become as large as desired, which means we never see the infinity.

When these results were compared to the measured results for large angles, they saw deviations. These deviations allowed Rutherford to estimate the size of the nucleus of the atom. Its radius is 100,000 times smaller than the size of the atom. This then ushered in the idea that the electrons must orbit a nucleus that contains most of the mass and all of the positive charge. But, as we will see next, there is a problem with this model.

When a particle orbits another particle, like the earth orbiting the sun, the orbiting object (earth) is constantly accelerating. But accelerating charged particles emit light, as you learned in your electromagnetism class. As they emit light, they must lose energy, and eventually spiral into the center. How long will this take? Problem 2.2 goes through this analysis for a hydrogen atom. The lifetime is extremely short. This indicates a severe problem with such a model. Niels Bohr (see Fig. 2.4) took up this problem [1]. He had a bold idea—assume that certain orbits were stable and would not radiate light as classical mechanics says they should. The emitted light then emerges from transitions between these different energy levels. This last point is one that was long known in spectroscopy—the emitted lines could be explained as being differences of a fixed set of energies, called the Rydberg-Ritz combination principle. The emerging idea was that quantum mechanics is a *restriction* on classical mechanics. Here, the restriction is that only classical orbits that are quantized in some specific fashion are allowed. Problem 2.3 explains how this theory worked for circular orbits.

Bohr, and later Arnold Sommerfeld [9] (see Fig. 2.5), worked out the rules for how to restrict classical mechanics into quantum mechanics. For periodic motion, this required an integral of the area of the orbit in phase space (a space characterized by the position and momentum of the particle at each moment in time) to be an integer multiple of Planck's constant. Using this so-called Bohr-Sommerfeld quantization rule, Sommerfeld was able to generalize Bohr's ideas from circular orbits to ellipses



Fig. 2.4: Niels Bohr in the early 1920's. Bohr developed the first quantum theory for the atom, based on a set of *ad hoc* postulates that worked remarkably well for determining the spectra of hydrogen. He won the Nobel prize in 1922. [Library of Congress, George Bain Collection, LC-B2- 5894-8. Public domain. Modified to look like a stencil by the authors.]



Fig. 2.5: Arnold Sommerfeld in 1923. Sommerfeld was involved in many aspects of developing quantum mechanics. Although he never won the Nobel prize himself, he helped train many who did including Werner Heisenberg and Wolfgang Pauli. [Transocean Berlin, G.m.b.H., courtesy of AIP Emilio Segrè Visual Archives, W. F. Meggers Collection. Public domain. The authors converted it into a stencil.]

and was even able to solve them in the relativistic case. Sommerfeld's quantized energies agreed exactly with the quantum results later found by Dirac for relativistic hydrogen. There seemed to be something remarkable about this *ad hoc* approach. You will investigate it for the harmonic oscillator in Problem 2.4.

But it had issues too. When applied to helium instead of hydrogen, it failed. There was no clear way to make it work for nonperiodic motion. And so on. Many physicists did not like this arbitrary quantization of classical orbits with no clear reason why. There were some ideas, such as Ehrenfest's adiabatic invariants that motivated the approach, but, it ended up being a hodge-podge of ideas with no central theme to it. Otto Stern decided he wanted to test it to see if it was real. He wanted to directly observe the quantized classical orbits. He devised an experiment to do just this, and enlisted the aid of Walter Gerlach to carry out the experimental work. He got the support of Max Born and financial donations from Henry Goldman of Goldman-Sachs to carry out the work. This experiment is our next story, as it leads us directly into a better understanding of the quantum world, even though it took some time after it was completed to fully appreciate what the experiment really measured.

2.2 The Stern-Gerlach experiment

In the Bohr-Sommerfeld quantization scheme, the prediction is that only certain orbits are allowed (called space quantization). When one takes into account angular momentum, the prediction is that angular momentum states with integer multiples of $\hbar$ are allowed. Now, when an electron orbits the nucleus, it acts like a current loop, so we need to understand how a current loop behaves in a magnetic field. Usually dynamics in magnetic fields is a topic that is supposed to be in an introductory electromagnetism class, but often one does not get to it before the semester ends. So, we will carefully walk you through the material.

Consider a circular current loop oriented at an angle relative to a magnetic field. What are the forces and torques on the current loop (we assume it is floating and free to move). We first have to recall that current loops generate magnetic fields and the north pole can be found by the right hand rule—curl the fingers to go in the direction of the current flow and the thumb points in the direction of the north pole of the field. Recall that a magnetic field pushes the north pole of a magnet in the direction of the field and the south pole opposite to the field. Now, if we place this in a uniform magnetic field, you might think it will act just to align the field of the loop the same as the field of the magnet, but because the current loop has moving charges in it, the current loop acts analogous to a spinning bicycle wheel or a gyroscope and it instead precesses. Precession is a complicated motion. If we focus on the line perpendicular to the current loop and moving through the center of the loop, the tip traces out a cone shape with the angle of the line unchanged during the motion. This says the tilt of the current loop remains the same during the precession motion.

The next question we need to ask is what happens when the magnetic field is nonuniform? Since the current loop acts like an effective magnet, and the north pole

of a magnetic is pushed in the direction of the field, while the south pole is pushed opposite to the field, the magnet will be pushed in the direction where the net push is, taking into account the different forces at the two poles. So, an effective magnet with its north pole pointing up in a field that points up and increases going up will be pushed up. If it increases going down (or the magnet orientation is reversed), then the magnet will be pulled down. This allows us to separate the magnets according to their orientation and how the field changes in space. The fact that the angle of the effective magnet for the current loop remains unchanged during precession means that when we put a current loop into a nonuniform field, we have a constant push up or down.

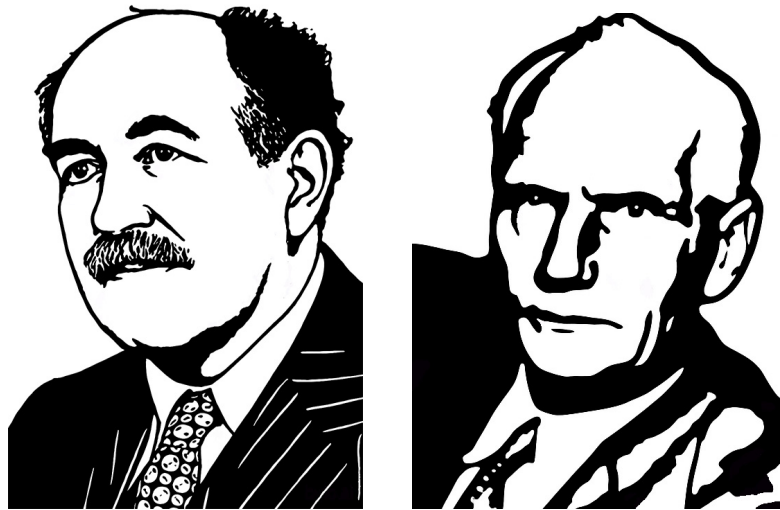


Fig. 2.6: (Left) Otto Stern in the 1940s and (Right) Walter Gerlach in the 1950s. The Stern-Gerlach experiment is one of the most important experiments in quantum mechanics even though it was originally misinterpreted in what it was actually measuring. Stern won the Nobel prize in 1943. Gerlach did not because of his work with the Nazis during World War II. [Stern image is public domain. Gerlach image ©Springer-Verlag from Z. Phys. 155 Iss. 4, 1 (1959) courtesy of AIP Emilio Segrè Visual Archives, E. Scott Barr Collection. Both are converted to stencils by the authors.]

Knowing all of this allows us to see that we can use an inhomogeneous magnetic field to separate atoms according to their angular momentum. Using the Bohr-Sommerfeld quantization condition, one expects the angular momentum to come in multiples of $\hbar$. This then allows one to see the effect of the quantized orbits on the motion of atoms. Otto Stern had this idea for an experiment fully developed in 1921 and he recruited Walter Gerlach to carry out the experiment with him at the

University of Frankfurt [5]. Stern had been working on experiments with atomic beams of silver and Gerlach was a superb experimentalist, so the idea to work together was a natural one. The challenge was to get the beams narrow enough that the separation between them could be seen. They also needed to amplify their signal so they could observe it. We will describe those details in a moment. But first, we need to explain more carefully just how the experiment works. This requires us to get deep into the equations that describe electromagnetism. Let us roll up our sleeves and get to work!

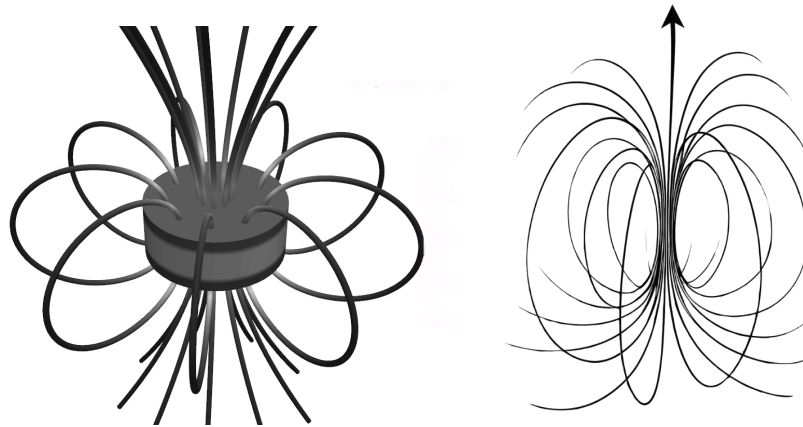


Fig. 2.7: Two views of magnetic field lines for an inhomogeneous field. (Left) Schematic of the magnetic field lines (solid line curves) above the north pole of a magnet. Because the field decreases as we go away from the poles, we can see the spacing increase, but we also see that the changes occur in more than just one direction. (Right) An expanded view of field lines showing they change in more than one direction. [Left image created with Magpylib [7]. Right image created by the authors.]

The first thing we need to do is to work with the structure of a magnetic field. As we already described, the experiment requires us to work with an inhomogeneous magnetic field meaning it will change in at least one direction. Because there are no magnetic monopoles, magnetic fields are divergenceless, so, we have $\vec{\nabla} \cdot \vec{B} = 0$, which is one of the Maxwell equations in differential form. For example, if we look at the magnetic field lines above the north pole of a magnet, they appear as shown in Fig. 2.7. You can see they are changing in more than one direction. We will use this fact in our analysis in just a moment.

The next thing we need to do is to consider the interaction of an electron orbiting the nucleus of an atom (in an orbit that Bohr says will be stable) and determine how it interacts with a magnetic field. One rule you should remember from electromagnetism is that we can imagine it to be similar to a current loop, where we use a right

hand rule to determine the direction of the effective north pole of the current loop (which we call the effective needle of the current loop). This procedure is illustrated in Fig. 2.8. You curl your fingers in the direction of the current flow and the thumb points in the direction of the north pole.

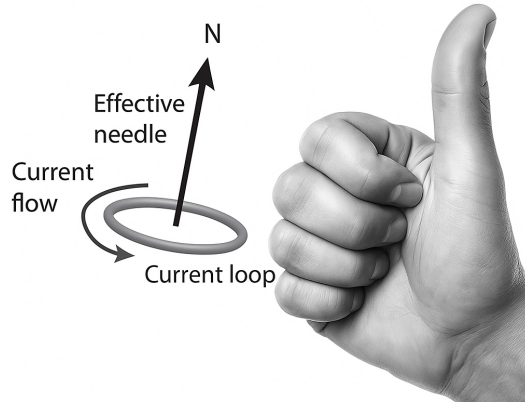


Fig. 2.8: Application of the right hand rule for a current loop. The fingers are curled in the direction of the current flow along the current loop and the thumb points in the direction of the effective north pole of the current loop.

Okay. Now that we have gotten those preliminaries done, we need to get quantitative and analyze a number of key situations. First, let us use the Lorentz force law for a vertical magnetic field, and a particle orbiting in a circle that is oriented at an angle of θ to the vertical. What is the net force on the electron from the magnetic field? The time averaged force is zero. You can see this by picking a point on the orbit, finding the velocity, and computing the Lorentz force, given by $-e\vec{v} \times \vec{B}$. But, if we wait until a later time, where the electron is at the opposite point of the circle, then the velocity is opposite, so the force becomes $e\vec{v} \times \vec{B}$. Hence, adding the two together, as we would in a time average, shows they will cancel. Since this holds for every point on the circular orbit, we find the net force is zero.

The next thing we look at is the torque. Taking the Lorentz force law and computing $\vec{r} \times \vec{F}$ with respect to the center of the loop, we find there is a nonzero torque, because the forces are opposite, at opposite points on the current loop, meaning they add together for the torque, if they are nonzero. Adding them together for a time average over the orbit leads to a net torque that is perpendicular to the angular momentum, which lies in the same direction as the “effective needle,” or your thumb using the right-hand rule we described above. When a torque is perpendicular to the angular momentum, the motion is precession, where the tip of the angular momentum vector rotates in a circle centered at the tail and perpendicular to the magnetic field. Problem 2.5 performs a semi-quantitative calculation of this result.

So, when a current loop, or an electron orbiting the nucleus of an atom, is placed in a uniform magnetic field, it will precess. If we want to apply a force to it, we need to place it in a nonuniform magnetic field. In this case, we will still see the atom precess, but, in addition, we will see a force which is determined by whether the force is larger on the north pole or the south pole of the effective needle.

We summarize this behavior with the definition of a concept called a magnetic moment. The magnetic moment is defined so that the force is equal to $\vec{F} = \vec{\nabla}(\vec{\mu} \cdot \vec{B})$ and the torque is $\vec{\mu} \times \vec{B}$. Here, for a particle in a circular orbit, the magnetic moment is proportional to the angular momentum and satisfies

$$\vec{\mu} = \frac{gq}{2m} \vec{L}, \quad (2.10)$$

where q is the charge of the particle in orbit, m is its mass, $\vec{L}$ is the angular momentum (with $|\vec{L}| = mvr$), and g is an additional factor called the g -factor. Classically, $g = 1$, but for many particles, g takes on different values. In fact, some neutral particles have $gq \neq 0$, such as the neutron!

Now that we know all of the behavior of the magnetic field of an electron orbiting in an atom, we are ready to describe and analyze the Stern-Gerlach experiment. Stern wanted to directly observe these orbits in the atom, which were quantized to only specific allowed orbits (with different angles for the circular orbits); note that we have not fully explored the origin of this quantization here, but it comes from the Bohr-Sommerfeld quantization rules and the quantization of angular momentum. His idea was that if the effective needle is pointing up for some atoms and down for others, then we can use an inhomogeneous magnetic field to separate them. Stern had already been working on experiments with atomic silver beams, so the idea seemed simple—take the atomic beam and put it inside a strong inhomogeneous magnet. Then collect enough atoms on a glass plate that they can be observed once the silver is oxidized to become a black color. Easy as this may sound, it is very hard to accomplish, because the beam size must be smaller than the splitting to be able to see the splitting. Because the splitting is small, the beam size also had to be small. Gerlach accomplished this by collimating the beam. This means he had it pass through three aligned slits in three different plates placed in front of the output of the oven that boiled off silver atoms. Imagine doing this when you cannot see the atomic beam! And, of course, it all had to be inside a vacuum as well. Definitely a hard task! Eventually Gerlach succeeded, just as they were getting ready to quit (because the money had run out). We will show you the results shortly. But first, we need to go through the full quantitative analysis of this experiment using classical reasoning. To do this, we follow the ideas of Devereux [2].

We begin with describing the magnetic field. At the center of the magnet, because the magnet is long in the y -direction, the field in that direction is essentially zero, except near the edges of the magnet, which we ignore. The field is then of the form

$$\vec{B}(x, y, z) = B_x(x, y, z) \vec{e}_x + B_z(x, y, z) \vec{e}_z, \quad (2.11)$$

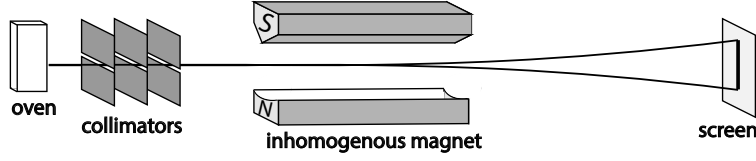


Fig. 2.9: Schematic of the classical Stern-Gerlach experiment with the beam direction being the y -direction, the vertical being the z -direction, and the direction out of the page being the x -direction. To the left is an oven that boils off an atomic beam of silver atoms. The atoms pass through three collimators before entering a strong inhomogeneous magnet, which separates the beam according to the projection of the effective needle on the vertical direction. The silver atoms hit a screen on the right, where they can be observed once enough hit the screen and the silver is oxidized to turn black in color. The atoms will hit all spots between the maximum positive projection of the magnetic moment on the vertical and the maximum negative projection, illustrated by the thick vertical line. Note that here, we illustrate a narrow circular beam, but the true beam in the experiment was thin in the vertical direction, but fatter in the direction out of the page, as you will see when we show the results below.

and we will be ignoring the y -dependence for the beam inside the long magnet. Because $\vec{\nabla} \cdot \vec{B} = 0$, we must have

$$\frac{\partial}{\partial x} B_x = -\frac{\partial}{\partial z} B_z, \quad (2.12)$$

because $B_y = 0$. Furthermore, because the curl of the magnetic field also vanishes, $\vec{\nabla} \times \vec{B} = 0$, and because $B_y = 0$, we also have that

$$\frac{\partial}{\partial x} B_z = \frac{\partial}{\partial z} B_x. \quad (2.13)$$

Finally, because the magnet is symmetric in the x direction, the field in the z -direction must be an even function of x about the center of the magnet, so that

$$\left. \frac{\partial}{\partial x} B_z \right|_{x=0} = 0. \quad (2.14)$$

Now, we recall that the force is $\vec{F} = \vec{\nabla}(\vec{\mu} \cdot \vec{B})$. This takes the form

$$\begin{aligned} \vec{F} = & \left(\mu_x \frac{\partial}{\partial x} B_x + \mu_y \frac{\partial}{\partial x} B_y + \mu_z \frac{\partial}{\partial x} B_z \right) \vec{e}_x + \left(\mu_x \frac{\partial}{\partial y} B_x + \mu_y \frac{\partial}{\partial y} B_y + \mu_z \frac{\partial}{\partial y} B_z \right) \vec{e}_y \\ & + \left(\mu_x \frac{\partial}{\partial z} B_x + \mu_y \frac{\partial}{\partial z} B_y + \mu_z \frac{\partial}{\partial z} B_z \right) \vec{e}_z = -\mu_x \frac{\partial}{\partial z} B_z \vec{e}_x + \mu_z \frac{\partial}{\partial z} B_z \vec{e}_z. \end{aligned} \quad (2.15)$$

The simplification comes from using the facts we learned above about the magnetic field.

Now that we know what the magnetic field looks like, we next examine the forces and torques on the atom. To do this, we need to first determine the magnetic moment. We use the approach followed in an electromagnetism class—the magnitude of the magnetic moment is given by the current flowing in the loop multiplied by its area. For an electron orbiting in a circle, the current is the charge per unit time flowing along the circle. This is given by $i = -ev/(2\pi r)$ and the area is πr^2 , so that

$$|\vec{\mu}| = \frac{emvr}{2m} = \frac{eL}{2m}, \quad (2.16)$$

where $L = mvr$ is the orbital angular momentum. Note that for a classical calculation, we choose $g = 1$.

Now, the motion is divided into two separate pieces. First, we will examine the precession motion of the magnetic moment assuming the field is uniform. Then once we know what $\vec{\mu}(t)$ is, we compute the force on the atom and use that to determine the final motion. Note that it is safe to assume the magnetic field is a constant for the torque calculation, because the torque is nonzero in the uniform field, and we assume the field is large, but its gradient is less so because the field is nonzero and upward in the entire magnet.

For the torque, we learned that $\tau = \vec{\mu} \times \vec{B}$, hence $\dot{\vec{L}} = \vec{\mu} \times \vec{B}$ or

$$\dot{\vec{\mu}} = -\frac{e}{2m} \vec{\mu} \times \vec{B}. \quad (2.17)$$

Defining the Larmor frequency to be $eB/2m$, we find that

$$\frac{d}{dt}\mu_x = -\omega\mu_y \quad \text{and} \quad \frac{d}{dt}\mu_y = \omega\mu_x. \quad (2.18)$$

These are two coupled linear differential equations. You can see the derivative of μ_x is proportional to μ_y and *vice versa*. How do we solve such equations, even if you have little experience with differential equations? Think about it for a moment.

The answer is differentiate the first equation, and use the second to decouple them. Namely, we have

$$\frac{d^2}{dt^2}\mu_x = -\omega \frac{d}{dt}\mu_y = -\omega^2\mu_x. \quad (2.19)$$

This differential equation asks the question, what function has its second derivative proportional to the negative of that function. From your calculus class, you should recall it is a sine or a cosine. So, we have that

$$\mu_x(t) = A \cos \omega t + B \sin \omega t. \quad (2.20)$$

How do we find the constants? By using the initial conditions. We have $\mu_x(t=0) = \mu_{x0}$ and $\dot{\mu}_x(t=0) = -\omega\mu_{y0}$, with μ_{x0} the initial value of the x -component of the angular momentum and similarly for μ_{y0} . This yields

$$\mu_x(t) = \mu_{x0} \cos \omega t - \mu_{y0} \sin \omega t, \quad \mu_y(t) = \mu_y \cos \omega t + \mu_x \sin \omega t, \quad (2.21)$$

and $\mu_z(t) = \mu_{z0}$.

Now that we have the magnetic moment as a function of time, we work on the force. We have

$$\ddot{x} = (-\mu_x \cos \omega t + \mu_y \sin \omega t) \frac{\partial B_z}{\partial z}, \quad \ddot{y} = 0, \quad \text{and} \quad \ddot{z} = \mu_{z0} \frac{\partial B_z}{\partial z}. \quad (2.22)$$

The force in the x -direction is oscillatory, and will average to zero. The force in the y -direction is always zero. The force in the z direction is constant and proportional to the value of the magnetic moment along the z -axis. One can see that this will cause the atoms to separate according to the projection of the magnetic moment along the z -direction, namely μ_{z0} . In Problem 2.6, you will put in quantitative numbers to verify these statements and to see how large the separation tends to be in a real experiment. Note that this analysis assumed the gradient in the z -direction is approximately constant, which is a reasonable assumption.

The result of this classical analysis is that when we run the Stern-Gerlach experiment, we expect to see the atoms separated according to their projection, and because the oven should create random initial projections, we will see a continuous range of different values on the screen, as shown in Fig. 2.9. Now, Stern believed he would see something else. Using the idea of quantization of angular momentum from the Bohr model, we anticipate only a finite number of different projections are possible, so we should see the result of the different allowed projections only.

So what is the result? In the experiment, after they finally got it to work, they saw two results only. See the experimental results from the original pictures in Fig. 2.10. This measurement was surprising for two reasons. First, Stern did not believe in Bohr-Sommerfeld quantization, so it was surprising that his experiment appeared to confirm it. Second, the classical theory says one should always see an odd number of projections, but here we had an even number. How is that explained? We will answer the second question shortly, as it arises from a re-analysis of this experiment that was completed a few years after the experiment was reported in 1922. But first, we will examine the consequences of this experiment further.

One of the ways to interpret the results of the Stern-Gerlach experiment is the way they did—the quantization of angular momentum allowed only two possible orbits. But, if we think of this further, we see it is more complex than this. What would happen if we could rotate the device? It seems like we should be able to find a direction where the splitting of the initial projections lies in the same plane. Then we would see no splitting. But, no matter how the experiment is oriented, it always shows two splittings (when done with silver atoms) and the splitting is always exactly the same. There is no single direction the two initial magnetic moments can point in that would always show the same splitting. So, there is no simple classical explanation to this result. It is a quantum result, and it is different from what any classical analysis can produce.

We mentioned earlier that the Stern-Gerlach experiment was difficult to carry out. This is in part because it is hard to align atomic beams that cannot be seen

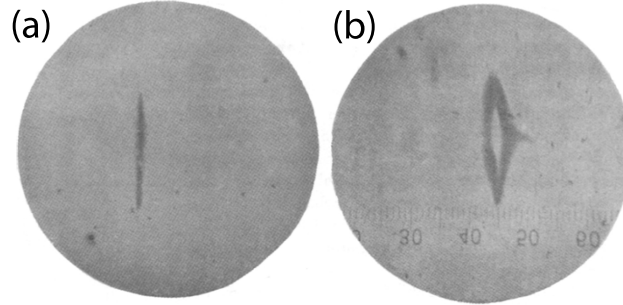


Fig. 2.10: Results of the Stern-Gerlach experiment from Ref. [5]. In panel (a) the magnetic field is zero, and in panel (b) it is on. Note that the lines separate in the center but join in the ends. This is because the field is inhomogeneous in the x -direction and becomes too small to separate the atomic beams when far enough from the center. [Figure © Springer-Verlag [5] and used with permission.]

and because even with a strong magnet, the separation is quite small. This means it is very difficult to do more complicated experiments like repeated Stern-Gerlach experiments. Yet, this has actually been done!

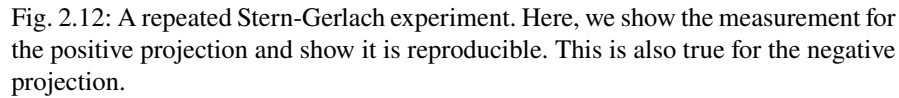
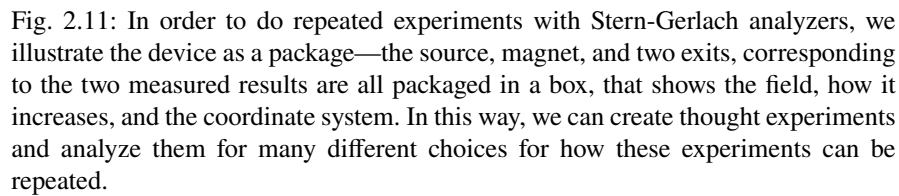
Now, when we find a curious new result, we want to do more experiments to learn more about it. Back in the 1930s, this was actually done, but the series of experiments we will discuss that help us understand how this works are a bit different from those experiments. We will assume our new experiment is so good, that we can observe the results a single atom at a time (even though this would be very challenging to do, because we always need the amplification step of any quantum measurement to be able to see individual atoms).

In Fig. 2.11, we show how we package the Stern-Gerlach experiment into something we call a Stern-Gerlach analyzer. This device illustrates the direction of the magnetic field and shows how it will separate the atoms according to their two different projections. We now use this device to create some different experiments. We want to point out that these experiments are really thought experiments. They have not been directly performed, although some experiments similar in style to this have been done and we believe all of these thought experiments are analyzed completely correctly.

Figure 2.12 illustrates that the separation is repeatable. If we measure a positive projection once and measure it again, we always see it to be positive. The same holds for the negative projection.

Figure 2.13, we show that if we measure the projection once, and then take one projection of the separated atoms and measure them on the opposite direction, we find the opposite projection, again as expected.

The last experiment we look at here is the case where we send the beam through a vertical analyzer, take the atoms with a positive projection and then place them through an analyzer in the horizontal direction. What we find is that half of the time



Historically, the idea that these results corresponded to the quantized Bohr-Sommerfeld orbits in an atom was soon viewed as an incorrect interpretation. In 1925, Goudsmit and Uhlenbeck (see Fig. 2.15) predicted that the electron carried an internal degree of freedom for angular momentum that corresponded to projections of $\pm\hbar/2$ [12]. They initially thought this occurred due to the electron being a sphere of charge whose size was the classical electron radius and it spun on its axis, like the

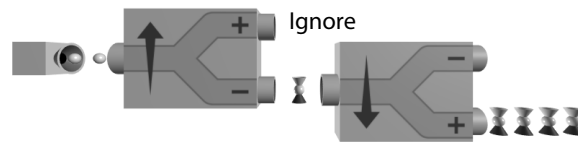


Fig. 2.13: A repeated Stern-Gerlach experiment. Here, we show the measurement for the negative projection and show it is reproducible when we measure it along the opposite oriented field.

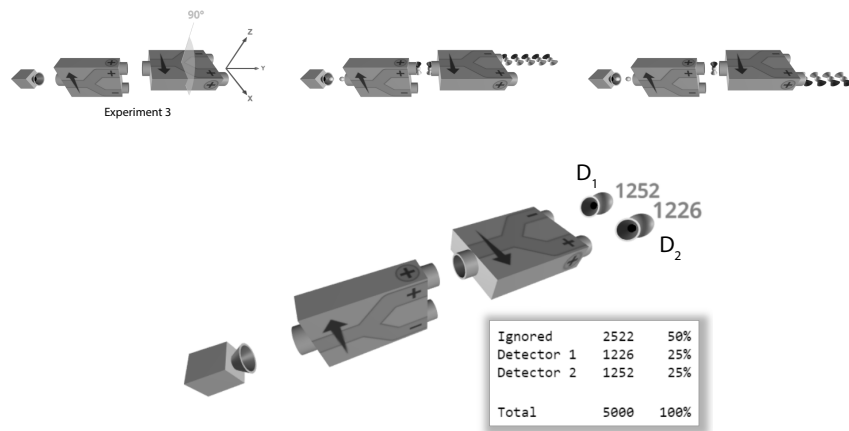


Fig. 2.14: Now we shoot the beam through a vertical analyzer and then take the positive projection and shoot through a horizontal analyzer. We find half of the time it has a positive projection and half of the time negative. We cannot tell which exit any given atom will take—we can only predict the probability after the experiment is repeated many times. The inset summarizes the result of a simulated experiment with single-atom detectors on each exit.

earth spins on its axis. This was quickly viewed as a problem, because the speed at the surface of this hypothetical sphere of charge would be moving at faster than the speed of light to create the angular momentum observed. Furthermore, the splitting seen by Stern and Gerlach were two times larger than this prediction by Goudsmit and Uhlenbeck. This latter result comes from the fact that $g = 2$ for an electron, a fact that was worked out by Lewellyn Thomas in 1926. The Stern-Gerlach experiment was properly reinterpreted as corresponding to a measurement of the spin projection in 1927. In an unfortunate turn of fate, Ralph Kronig had the same idea a few years before Goudsmit and Uhlenbeck, but he was working under Pauli at the time and

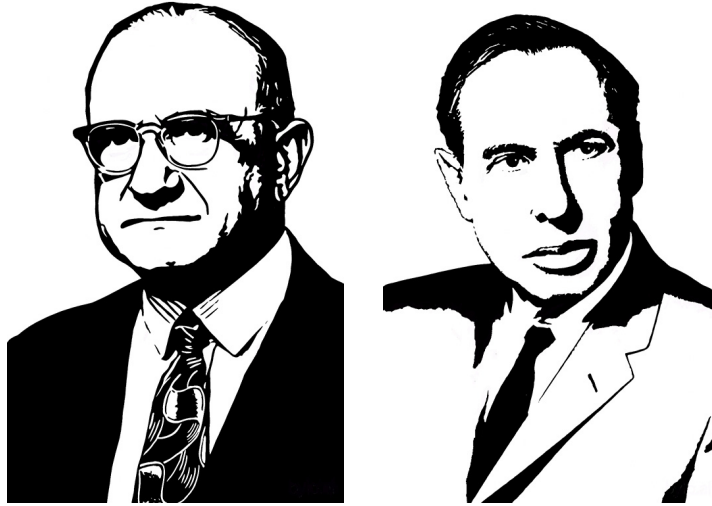


Fig. 2.15: Samuel Goudsmit (left) and Georget Uhlenbeck (right) in the 1950s. The pair discovered the spin of the electron. [Goudsmit image credit Electronic News Photo, courtesy AIP Emilio Segrè Visual Archives. Uhlenbeck image credit AIP Emilio Segrè Visual Archives, Physics Today Collection. Both images converted to stencils by the authors.]

Pauli vehemently opposed the idea, so he never published it. Pauli continued to be adverse to the idea for a number of years, but then eventually accepted it when it did not need to be thought of as a spinning of charge. In the problems you will investigate the classical radius and the velocity at the equator for the spinning ball of charge.

The Stern-Gerlach experiment, like many of the other early experiments is important, in part, because it yields to a nice and simple classical analysis, but when we compare that to the experimental results, we see they do not agree. This tells us something beyond classical physics is going on. We find that the results are not deterministic, but rather are probabilistic. This means the whole idea of determinism comes into question. Our perspective is different however. We feel that understanding the classical picture is key to understanding the more intriguing quantum result. We will develop the consequences of this quantum result soon. At this point, we simply marvel at its beauty, even in this raw and unexplained form.

2.3 The analyzer loop

We start this section by discussing the rule for what the probability is when we examine atoms exiting a vertical analyzer from the positive exit and the second analyzer is at an angle θ relative to the vertical. The probability to be measured

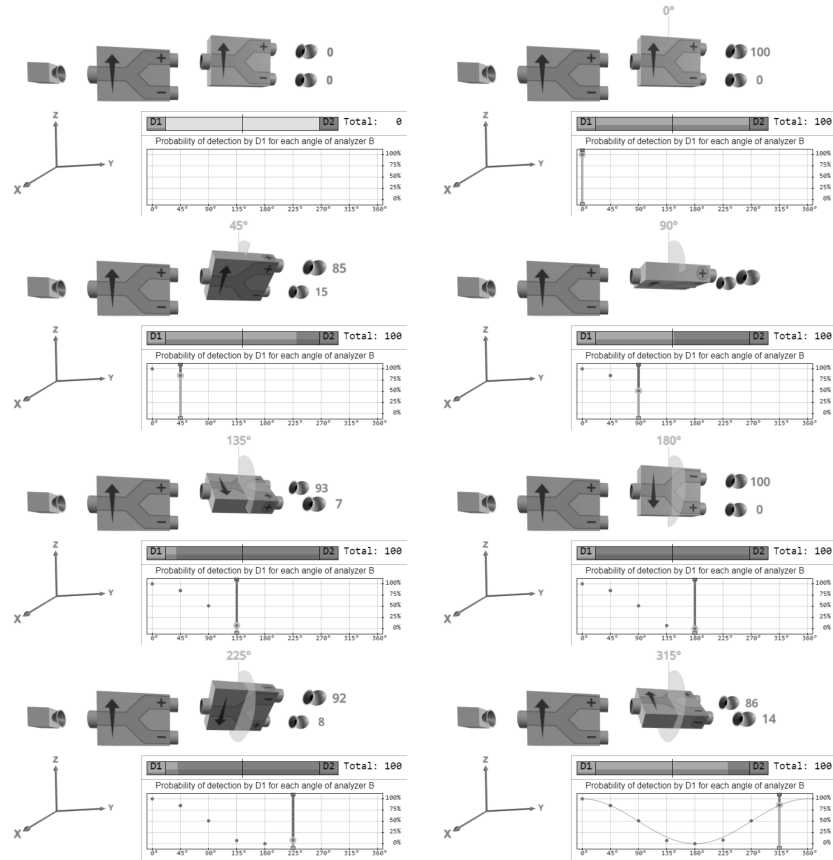


Fig. 2.16: This figure summarizes what the behavior is as the angle between the two analyzers increases from 0 to 360°. The last plot shows the complete curve, which has a probability to measure a positive projection given by $\cos^2 \frac{\theta}{2}$.

with a positive projection is $P_+ = \cos^2 \frac{\theta}{2}$ and the probability to be measured with a negative probability is $P_- = \sin^2 \frac{\theta}{2}$. At this point, we are just telling you what this is. In Chap. 3, we will derive it for you. Having such a definite formula now gives us predictive power. The formula holds for other geometries as well as long as θ is the relative angle between the first and second analyzer.

This summarizes the basic behavior of quantum systems, which is that they are governed by an intrinsic randomness. Note that random does not mean totally arbitrary. Instead, it means that for a given experiment that has multiple possible results and for any given run (or sometimes called shot) of the experiment, we cannot predict precisely what will happen. All we can do is predict the probability of how the measurements will be distributed, if they are repeated many times. This randomness is a fundamental property of the quantum world. It may make you

feel uneasy, but there are many things you have already seen in your life that are random—the result of flipping a coin is one of the most common examples that you are already familiar with. So, even though this random behavior may seem odd to you, it should not feel unfamiliar. You have seen many examples of randomness in your life already.

You may have heard that quantum mechanics is the most accurate scientific theory in all of science. This is true. But, how can that be reconciled with the notion of randomness? It turns out many quantum predictions can be made that involve no randomness whatsoever, such as the energy of the ground state of hydrogen. So, while quantum mechanics has an intrinsic randomness to it, not everything in quantum mechanics is random. And we can predict the probabilities of the random events very accurately, so randomness does not make the theory problematic. It just makes it different from many theories you have experienced so far. Sometimes we use the word stochastic instead of random so as to not bring in any notion that we cannot make predictions due to the intrinsic randomness.



Fig. 2.17: Paul Dirac in the 1920s. Dirac was one of the founders of quantum mechanics and won the Nobel prize in 1933 for this work. He created the most powerful notation for quantum mechanics, which is appropriately called Dirac notation. [Photograph by A. Bortzells Tryckeri, courtesy AIP Emilio Segrè Visual Archives, E. Scott Barr Collection, Weber Collection, and in the public domain. Converted to look like a stencil by the authors.]

We also want to start to develop some formalism to go with our theory. In quantum mechanics this is done with a notation named after the Nobel-prize-winning physicist Paul Dirac [3], shown in Fig. 2.17. This notation is both straightforward and also

very powerful. Let us see how it works and how we can apply it to the Stern-Gerlach experiment.

In the Stern-Gerlach experiment, we found we had two states—a positive projection state and a negative projection state. This remains true in more complex systems as well—we always need to describe quantum states. We abstractly represent these quantum states in terms of what we call quantum state vectors. We need to learn a lot about how we work with quantum states, and that is coming, but for now, we have to decide on the notation we wish to use.

In your introductory physics class, your instructor probably spent a fair amount of time discussing how the vector you use in three dimensions to describe position, or momentum, or force, is an abstract mathematical object and is not just the three numbers used to represent it in a specific coordinate system. We represented these vectors with boldface or overhead arrows, like $\mathbf{F} = \vec{F}$ for the force. You got used to this notation fairly quickly. Things are more complicated in quantum mechanics though, because the vectors can live in two-dimensional, or three-dimensional, or even infinite-dimensional spaces. So, in 1939, Dirac invented a compact way of representing quantum state vectors and other quantum objects that we will encounter soon [3].

Sounds cool, right? Well it is!

Before telling you about what this notation is, let's first think about how we might do it. What we want is a simple way to represent vectors, and, at this point, we will also want a simple way to represent the inner product between two vectors.

So what do we need to describe a vector? We just need its name. We also want to graphically represent inner products (a generalization of the three-dimensional dot product to arbitrary dimensions that also allows the vectors to be composed of complex numbers). If I had a vector, which I represent as a column vector, and I want to take its inner product with another column vector, what I need to do is to first transpose the column vector into a row vector, then I need to take the complex conjugate of all of the coordinates of the row vector, and finally, I multiply each coordinate for each dimension of the row and column vector together and add then together. In equations, we have the inner product between $\vec{a}$ and $\vec{b}$ is

$$\vec{a} \cdot \vec{b} = \sum_{n=1}^N a_n^* b_n, \quad (2.23)$$

for an N -dimensional vector. Note that this notation is not very good, because we need to make the a -vector a row vector and take its complex conjugate. Dirac's solution is much better. We enclose the name of the vector inside a special symbol, and call it a ket. In this case we have $\vec{b} \equiv |b\rangle$, and this “ket” is a column vector. The row vector (with its complex conjugate taken) is called a “bra” and is denoted by $\langle a|$. Then the inner product is called a bra-ket or bracket and is given by $\langle a|b\rangle$. That is all there is to it for vectors and inner products!

Using this notation, we describe a state with a positive projection as $\uparrow$ and of a negative projection as $\downarrow$. Then we use a subscript to denote the direction. So a positive projection atomic state in the vertical or z -direction is $|\uparrow\rangle_z$ and a negative

projection atom in the horizontal or x -direction is $|\downarrow\rangle_x$. We have not yet discussed why we might want to take the inner product between two quantum states, but we will, and soon. I hope you will agree it really is a powerful way to do calculations, as we develop it further.

Now, we go back to the Stern-Gerlach experiment. We discussed at the beginning of this section how to determine the probability for which exit the atom will emerge if the second analyzer is at a different angle from the first. What happens if we add a third analyzer? Does the atom remember its original state, or does it depend on just what the state was as it enters each analyzer? From the way we stated our probability rule, you should see the result of an experiment, where an atom enters an analyzer depends solely on the state it has when it enters. This means the atoms have no memory of the past, they only know what they currently are. We colloquially describe this as “the atom is stupid!” Keep this notion in mind whenever you analyze an experiment. In addition, because the atom is stupid, but you are not, you can always analyze these types of experiments, even complex ones.

We now introduce a new apparatus that we name the Stern-Gerlach analyzer loop, which was originally discussed by Eugene Wigner [13]. This device erases the measurement of the Stern-Gerlach analyzer. It consists of two pieces. The front end is the conventional Stern-Gerlach analyzer, while the back end is a new device that undoes what the first analyzer did. It brings the two paths back to one in such a way that we cannot tell which path the atom traveled on as it went through the device. Due to this fact, the measurement of the projection of the effective needle is erased, and the atom proceeds as if the analyzer loop was not there. In other words, the analyzer loop does nothing.

At first you might say, why would anyone want this? But, you will discover that if we allow ourselves to do some experiments on the atoms when they are separated (including measurements), then the analyzer loop allows us to create quite complex experiments. We will be able to do some mind-blowing things with them.

Note, however, that no one has made such an analyzer loop from conventional magnets and atomic beams. Never. This may make you concerned about how we can be confident we know what happens when we examine the results of different thought experiments made from them. And we do not blame you if you feel that way. It turns out that these analyzer loops have been made by more advanced technology involving trapped atoms, and they perform exactly as we have described them [6]. So, you have no need to worry about these devices actually working. They really do.

The construction of an analyzer loop is straightforward. It starts with a Stern-Gerlach analyzer that will separate an atom according to the projection of its magnetic moment along the direction the analyzer is oriented. We then place another device, which takes the two separated paths of the analyzer, and slowly joins them back together until the atom emerges from a single path. The second half undoes what the first half did, and does so in such a fashion that the projection that the atom has when it emerges is identical to the projection it had when it entered. We lose all information about which path the atoms took through the device. Such a device does nothing! The atom exits in the same state that it entered.

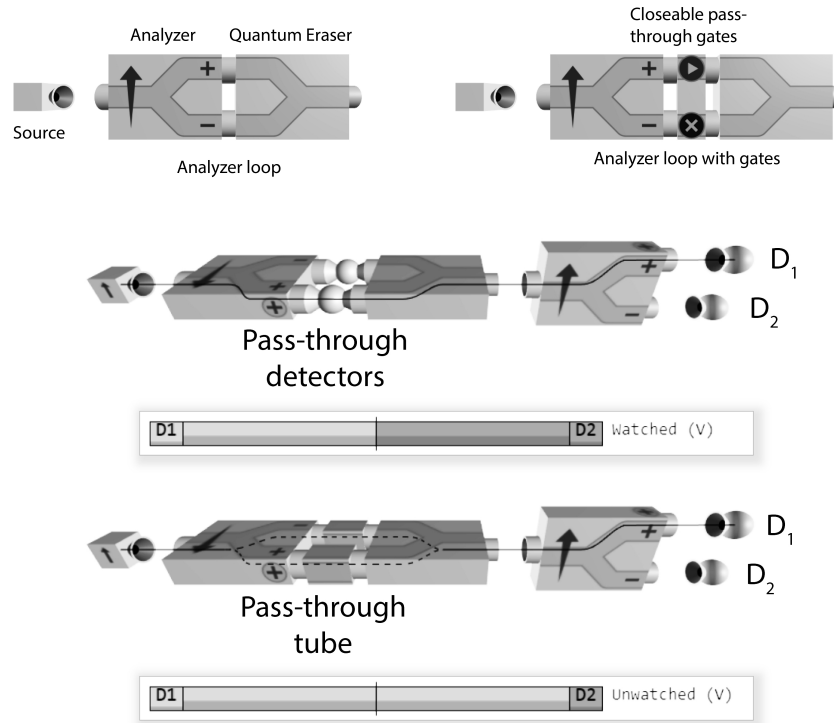


Fig. 2.18: (Top left) Schematic of an analyzer loop, with a source creating an atomic beam which enters an analyzer, and then goes through the second piece, called the quantum eraser, that undoes what the first part did. (Top right) An analyzer loop with closable gates. The top one with an arrow allows atoms through, while the bottom one with an x does not. (Middle) Now we use a source that passes through a vertical analyzer and takes only those with a positive projection, which we call an $\uparrow$ source. This analyzer loop has pass-through detectors, so the state of the atom changes when measured to being positive or negative along the x -axis. Hence, we see half to have a positive z -projection and half to have a negative projection. (Bottom) A similar arrangement as the middle has, but here with pass-through tubes, which do not perform a measurement and cannot determine which path the atoms take. Now, all atoms emerge with the same state they entered—here, it is positive along the z -axis.

To do something interesting with this, we create two gates in the device, when the atomic paths are split the farthest from each other. We can create an obstruction that the atom will hit if it goes on that branch of the path. Placing the obstruction on one of the paths is the same as measuring the projection on the path that was not blocked. Now the atom will emerge with a projection given by the path it took, the one with no obstruction, and it will lose all memory of its initial state; that is, if it did not hit the obstruction. We can do more complex things as well. We can add pass-through detectors that watch whether an atom passes one path or the other (or both if it can do that). We also will find it useful to have pass-through tubes as well, where the atoms can move through with no obstruction, but we are not watching them as they move. One can think of other things that we can do, but we will content ourselves with just these options. The analyzer loop is illustrated in Fig. 2.18, showing all of the options we will be working with.

Let us focus here on the cases with pass-through detectors or pass-through tubes. In the former case, we measure which path the atom goes on, so we know that it has a positive or negative projection on the x -axis while it is inside the analyzer loop. This means it enters the final analyzer with a definite projection on the x -axis. Regardless of which projection this is, we will find half of the time they emerge with a positive projection on the z -axis and half of the time a negative projection. This seems very reasonable. They started positive in z , but when they are measured, they change to $\pm$ on the x -axis. So, when they are measured at the end along the vertical, we use our rule to find the probability and see it is 50–50. Now, consider the bottom case. Here, our description gets more complicated. We know that the atoms will exit in the same state they entered, namely positive projection on the z -axis, so all atoms exit the positive exit of the trailing vertical analyzer. How do they do this? They cannot go through one path or the other, as that would be the same as the previous case, and we would have a 50-50 split. The atom cannot separate into two pieces and half goes on one path and half on the other. We cannot split atoms. So, how does it work? The answer is we do not know, and this is called the quantum mystery. Rather than try to explain it, we change our language and distinguish this case by saying the atom travels in superposition through the two arms, or, we say the state of the atom inside the analyzer loop is indeterminate. You may not like this response. It is uncomfortable. But, it is the best we can do. For, you see once we added the pass through detectors to see which path it took, we found it only took one or the other. But, the results of the final experiment changed! Nature does not allow us to see how the atom travels in superposition. It is frustrating, but it is the way things are.

You might say, let us get clever, and add one pass through detector on the front path and one pass-through tube on the rear path. What happens in this case? We get the same result as if we had two detectors. Why? Because if the detector registers an atom, it went through the $+x$ path. If it does not, it went through the $-x$ path. There are no other options. We still have which-way information, which means the path taken by the atom is not indeterminate—we know exactly what it is. We simply have to accept that watched atoms act differently from unwatched atoms and if they move in an indeterminate state, they somehow make it through the apparatus without splitting in half and without traveling on one path or the other. No, we do not know

how they do it, but the results of the experiment make it clear that they really do do this. The notion of do we have which-way information or not, is our first example of the importance of quantum information, and how working with quantum information is different than working with conventional information. Indeed, this is true, and we will explore these ideas a lot throughout the book.

Even if we do not have words to describe the behavior, we can describe it using our Dirac notation. One key aspect we will find in going through this exercise is that we need to have our states normalized, because they are used to calculate the probabilities of different experiments and something always happens, so the sum of all probabilities must be one. Furthermore, we need to postulate the Born rule, which says that the probability to measure a particular state in an experiment is given by the modulus squared of the coefficient of that state in the superposition. This statement will become clear in a moment.

Let us start with our formal analysis of the experiment with two pass-through detectors. We think of these experiments as state preparation followed by a measurement. So, we start in the state $|\uparrow\rangle_z$ as we enter the analyzer loop. Then, if the front pass through detector registers, we will find the state is $|\uparrow\rangle_x$, and if the rear detector fires, it is $|\downarrow\rangle_x$. Whichever state it is, we enter the final analyzer with this state. For either case, we find the probability is 50% because the angle is 90° . This means we can think of the state $|\uparrow\rangle_x$ as an equal superposition of the $|\uparrow\rangle_z$ and the $|\downarrow\rangle_z$ states. To determine the full superposition, we need to know the relative complex phase of these coefficients. For now, we will tell you the answer, but you will be calculating them in the next chapter, so just bear with us for now. We have

$$|\uparrow\rangle_x = \frac{1}{\sqrt{2}} |\uparrow\rangle_z + \frac{1}{\sqrt{2}} |\downarrow\rangle_z \quad \text{and} \quad |\downarrow\rangle_x = \frac{1}{\sqrt{2}} |\uparrow\rangle_z - \frac{1}{\sqrt{2}} |\downarrow\rangle_z. \quad (2.24)$$

Note that these two quantum states are orthogonal, which can be seen if we take the inner product of each of them and note that ${}_z\langle\uparrow|\uparrow\rangle_z = {}_z\langle\downarrow|\downarrow\rangle_z = 1$ and ${}_z\langle\uparrow|\downarrow\rangle_z = {}_z\langle\downarrow|\uparrow\rangle_z = 0$. This states that the eigenstates are normalized and eigenstates with different eigenvalues (positive or negative projections along the same axis) are orthogonal. Again, we will establish these results fully in the next chapter. The expressions in Eq. (2.24) are superpositions. This explains the results of the first experiment. Note that the formulas in Eq. (2.24) aren't to be thought of as simple equalities. They are different ways of expressing the exact same quantum state—in other words, they illustrate two identical expressions.

Now, let us do the same for the pass-through tubes experiment. We start in the state $|\uparrow\rangle_z$ as we enter the analyzer loop. Then, as we go through the pass-through tubes, we move in superposition in the state $\frac{1}{\sqrt{2}} |\uparrow\rangle_x + \frac{1}{\sqrt{2}} |\downarrow\rangle_x$. Note that this state is identical to the state $|\uparrow\rangle_z$. Now, when we exit, we re-express the state in the z -basis as $|\uparrow\rangle_z$ and all of the atoms go to the positive exit.

So, we have a clear formalism to describe the experiment, and the formalism uses different methods for the two experiments, which leads to different final results. But, the formalism does not tell us what it means to exist in a superposition state, nor

does it tell us how the atom manages to do this. It turns out, for now, we do not know how to answer this question, and hence, this is why we call it the quantum mystery.



Fig. 2.19: Two perpendicular analyzer loops with pass-through gates that can be opened or closed. The source inputs atoms in the state $|\uparrow\rangle_z$. The experimental results for the final output state will vary according to which gates are open and which are closed.

We have one more experiment we want to describe in this section. It is a bit tedious to get through, so bear with us. The experiment is illustrated in Fig. 2.19. We have a polarized source inputting $|\uparrow\rangle_z$ atoms from the left and it enters a horizontal (A) and then a vertical (B) oriented analyzer-loop. We examine all possibilities for the gates to be open or closed. First, we have all gates open. In this case the analyzer loops do nothing, so when we go through the first analyzer loop, the path is indeterminate, and it exits in the $|\uparrow\rangle_z$ state. Then for the second analyzer loop, we travel on the positive- z branch and exit in the $|\uparrow\rangle_z$ state. All atoms that entered will exit. What if we close the $-x$ branch of the first analyzer? Then half of the atoms hit the blockage and are lost. For the other half, they take the positive- x branch and we emerge from the first analyzer loop in the $|\uparrow\rangle_x$ state. Then the atom path is indeterminate in the second analyzer loop and the atom emerges in the $|\uparrow\rangle_x$ state, because the second analyzer loop does nothing. Now, half of the atoms entering ultimately emerge at the end. The analysis is similar if we block the $+x$ branch, except here atoms take the negative- x branch and emerge in the $|\downarrow\rangle_x$ state, again with half of the atoms emerging. If instead, we close the positive- z branch of the second analyzer loop, nothing gets through. Finally, if we close the negative- z branch of the second analyzer loop, all atoms emerge in the $|\uparrow\rangle_z$ state.

Let us now examine what happens if we close two gates. In any case where we close two gates in the same analyzer loop, nothing gets through, so we will close them in different analyzer loops. This gives us four possibilities. We will carefully analyze the case where we close the negative- x and negative- z branches. The other cases are similar. Since the negative x branch is closed, we travel through the positive- x branch of the first analyzer loop and emerge in the $|\uparrow\rangle_x$ state. Then we travel on the positive- z branch of the second analyzer loop and emerge in the $|\uparrow\rangle_z$ state. Half of the atoms are lost in each analyzer loop, so we only have one-fourth of the atoms that originally entered exiting the final analyzer loop. Similar results occur with all

other combinations. One fourth of the atoms emerge and in the state of whatever the final open gate was in the second analyzer-loop. Closing three gates or four gates is not interesting, because in these cases all atoms are blocked.

You might not have realized it, because this analysis was rather dry. But, we discovered something truly amazing. Go back over our results and see if you can find out what it is.

Did you find it? If not, do not worry, as we will now explain what it is. Look at the case where only the positive- z gate is blocked. In that case, we had no atoms make it to the exit. Now, look at the case where we add in additional blockage in the first analyzer-loop. The case where both negative- x and positive- z are closed. In this case, we find one-fourth of the atoms make it though, all exiting in the $|\downarrow\rangle_z$ state.

You might still be saying “so what.” But now think carefully about what just happened. We added more blockage in the system, and the number of exiting atoms went up! Hence, we can tell whether someone added a blockage in the first analyzer just by seeing whether any atoms emerge. This atom does not interact with the blockage. It is the atom that gets through and we detect it. Hence, we learn that a gate in the first analyzer loop is closed, by detecting an atom that did not hit the blockage! How does it know there is a blockage! This type of experiment is called an interaction-free experiment, and we will see it again multiple times, eventually showing you how to see something without shining light on it, which is called “quantum seeing in the dark.”

We have one more thing to cover in this section before we end. We want to introduce the notion of entanglement. Entanglement is a special form of superposition, but it cannot be described using the states we have been working with. Instead, we need to use a new notation called a tensor product to describe entanglement. You will find that the notation for it is rather simple, so we can describe it pretty quickly.

Let us start back with the analyzer and suppose it is oriented vertically. We determine the projection of the magnetic moment on the z -axis by observing the atoms that emerge from the positive or negative exit of the analyzer. Note that it is the spatial position that we observe, and we infer the projection. This means there really is a second degree of freedom in these systems—the spatial position. We can describe it with kets given by $|\text{up}\rangle$ and $|\text{down}\rangle$ to denote the two different spatial positions. How do we describe the quantum state of the atom? Well you might think, just expand the name of the ket to $|\uparrow, \text{up}\rangle_z$, but if we do that, we lose the opportunity to modify each component of the state separately. Instead, we use the two separate states and join them with a tensor product symbol, so we have the state $|\uparrow\rangle_z \otimes |\text{up}\rangle$. We will find later that this approach works well formally, because it allows operators to act on the different states independently, which would be more complicated to do with the other notation. Mathematically, this notion of a tensor product is a well-defined object and there are rules for how to work with them. At the moment, we will only state one of them, namely how to take the inner product of two states. We use the same ordering for the bras as we did for the kets, and when taking the inner product, we have an inner product for each ket where we match the bras from the same space together. Finally the inner products for each of the degrees of freedom are multiplied together. So, we would have

$${}_z \langle \uparrow | \otimes \langle \text{up} | \cdot | \uparrow \rangle_z \otimes | \text{up} \rangle = {}_z \langle \uparrow | \uparrow \rangle_z \langle \text{up} | \text{up} \rangle . \quad (2.25)$$

Now that we have discussed these mathematical details, let us describe the quantum state in the middle of a horizontal analyzer-loop that has an $|\uparrow\rangle_z \otimes |\text{center}\rangle$ state entered into it (that is, a state that has a positive projection in the z direction and is spatially moving in the central beam as it enters the analyzer-loop input. In the middle of the analyzer-loop, it will be in the state

$$\frac{1}{\sqrt{2}} |\uparrow\rangle_x \otimes |\text{front}\rangle + \frac{1}{\sqrt{2}} |\downarrow\rangle_x \otimes |\text{rear}\rangle . \quad (2.26)$$

Now, you can see that this state cannot be factorized across the tensor product. It must be expanded in those two specific terms. Such a state is called an entangled state (if it could be factorized, it would be called a product state). One would have a product state, for example, if the spatial state was front for both, but, of course, this would not be a state we would have in the analyzer loop.

So, do we say that a typical state of the atom when it is in the interior of an analyzer loop is a superposition or is it an entangled state? The answer is that it depends on how detailed you wish to have the discussion. Because in the analyzer loop the spatial position and the projection of the magnetic moment are completely correlated with each other, we can do it either way. Later, we will find examples where we must work with entangled states, and we will find really cool behavior occurring in those cases.

2.4 Summary

We have done a lot in this chapter. We started by describing some of the history about how quantum mechanics was developed, especially the changing models of the atom as new experiments were performed which gave us more complete information about the atom. We focused much of our discussion on the Stern-Gerlach experiment, which is a truly quantum experiment, because the quantum result is different from the classical prediction. Using this experiment as an example allowed us to introduce many of the interesting ideas of quantum mechanics, both conceptually and formally.

Along the way, we already discovered a number of interesting behaviors. We were introduced to the quantum mystery, where we were forced to describe systems in terms of a superposition, but we do not have a clear picture of just how an atom moves in superposition. Furthermore, when we set up an experiment to try to find out, we see the experimental results change, corresponding to a system that is not in a superposition. This is frustrating. Nature refuses to reveal its secrets to us. We also described how one can explain the Stern-Gerlach experiment using entangled superpositions. You were introduced to the notion of an interaction-free experiment as well.

Many of the ideas developed in this chapter were influenced by Feynman's *Lectures on Physics* [4], and by Styer's *The Strange World of Quantum Mechanics* [10].

A lot of this stuff was cool, even if it may take some time to get used to. We will show many more even cooler things in the next few chapters, so get ready. We are in for an adventure!

Chapter 2 problems

Problem 2.1 Rutherford scattering. In the main text, we carefully derived the scattering formula for the Coulomb interaction. We described how deviations from this formula were used to estimate the size of the nucleus. Write a short essay explaining how one might go about first determining what the deviations are and second determining how the deviations lead to a size of the nucleus that is 100,000 times smaller than the atom. You may need to do some research on Rutherford's original analysis to determine how this could be done.

Problem 2.2 Lifetime of the classical atom. If we describe the hydrogen atom classically, which we will take, for simplicity, as moving in a circular orbit, then because the electron is accelerating and is a charged particle, it must radiate light. The power radiated is given by the Larmor formula, which often is not discussed in an introductory electromagnetism class. The formula is simple, but the derivation is not, so we will just use the formula here for this problem. It says the power radiated by an electron accelerating with acceleration a is

$$P = -\frac{2}{3} \frac{e^2 a^2}{4\pi\epsilon_0 c^3}. \quad (2.27)$$

We assume the decay of the orbit is slow enough that the acceleration is radial directed and comes from an approximately circular orbit. By comparing the Coulomb force to the required acceleration, you need to show that

$$\frac{e^2}{4\pi\epsilon_0 r^2} = \frac{mv^2}{r}. \quad (2.28)$$

Use this result to show that the total energy is

$$E = -\frac{r_c}{2r} mc^2, \quad (2.29)$$

where $r_c = e^2/4\pi\epsilon_0 mc^2$ is the classical radius of the electron (which is approximately 2.8×10^{-15} m). Using the Larmor formula, and computing the time rate of change of the energy, you should find that

$$r^2 \dot{r} = \frac{1}{3} \frac{dr^3}{dt} = -\frac{4}{3} r_c^2 c. \quad (2.30)$$

Solve this differential equation to determine the time it takes to fall from the Bohr radius $a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2} \approx 0.529 \times 10^{-10}$ m to $r = 0$. You should find the lifetime of the atom is on the order of 10^{-11} s.

Problem 2.3 The Bohr model. Bohr assumed that quantum mechanics was a restriction on classical mechanics. Rather than having an atom that decays rapidly by having the electron spiral to the nucleus, he assumed that the atom had stable orbits quantified by having the angular momentum of the orbit satisfy $mvr = n\hbar$ for $n = 1, 2, \dots$.

Equate the mass of the electron times the acceleration of the circular orbit of radius r to the Coulomb interaction for hydrogen and use the quantization condition on the angular momentum to determine a formula for the allowed radius of the circular orbit. Your answer will be a simple function of n multiplying the Bohr radius $a_0 = \frac{4\pi\epsilon_0\hbar^2}{me^2} \approx 0.529 \times 10^{-10}$ m.

Calculate the total energy as a function of n and a_0 to determine the different allowed energy levels.

Bohr further assumed that the spectral lines of hydrogen were formed by transitions from one n value to another. The energy of the light was given by $E(n_i, n_f) = E_{n_i} - E_{n_f} = 2\pi\hbar f$, with f the frequency of the emitted light. Show how to use this idea to derive the Balmer formula for the optical transitions (with $n_f = 2$)

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right) \quad (2.31)$$

for $n = 3, 4, \dots$. Bohr's ability to derive an explicit formula for R_H in terms of fundamental constants was a huge achievement of the time. Now you have repeated this achievement!

Problem 2.4 Bohr-Sommerfeld quantization. Consider the harmonic oscillator that has an initial extension of x_0 but no initial momentum. If the frequency of the harmonic oscillator is ω and the mass is m , then you should be able to show that the position satisfies

$$x(t) = x_0 \cos(\omega t). \quad (2.32)$$

Differentiate this expression with respect to time to construct $p(t)$. Then evaluate the integral

$$\int_{\text{period}} p dx = \int_0^T p(t) \dot{x}(t) dt, \quad (2.33)$$

and require the result to be equal to $2\pi n\hbar$; here T is the period of the motion. This gives restrictions to the allowed values of x_0 . Express x_0 in terms of n and then compute the energy. Your final result is what the energy of the harmonic oscillator was thought to be in the early 1920s. The answer is wrong and was corrected by Heisenberg in 1925. We will solve it a number of different ways later in the book.

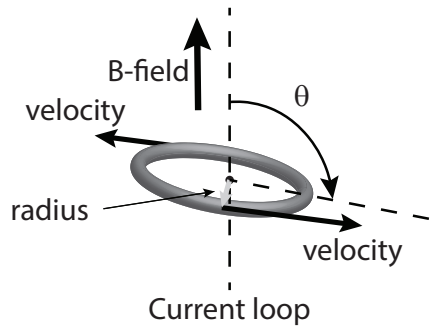


Fig. 2.20: Schematic of a current loop, which can be thought of as a particle of charge $-e$ moving around the circle with constant velocity v . The current loop is oriented at an angle of θ to the vertical and the magnetic field is vertical.

Problem 2.5 Forces and torques on a current loop in a constant magnetic field.

Fig. 2.20 shows the schematic of a current loop at an angle of θ to the vertical, where we wish to calculate the forces and torques on it. One can consider the current loop to consist of a single particle of charge $-e$ and moving with constant velocity v .

(a) Use the Lorentz force law for the two points on the current loop indicated in the figure and verify that the forces are equal in magnitude and opposite in direction. Further argue that the net force must vanish.

(b) Consider the torque for the two positions shown in the figure. You should be able to verify that these two torques vanish, because the force is in the same direction as the radius at these two locations.

(c) Now consider the points 90° out of phase, where the velocities are into and out of the page. You should be able to verify that these torques do not vanish, and are proportional to $\cos \theta$ and directed into the page, after taking into account that the electric charge is negative.

(d) Consider other pairs of points on the circle. Does the torque vanish or not? What direction does it point in if it does not vanish?

(e) Since the integral of the torque over the whole current loop, is just the sum of each of these pairs of torques, explain whether the torque vanishes, or is nonzero. If nonzero, what direction is it?

(f) In the text, we said the magnetic moment is constructed so that the total torque is given by $\vec{\tau} = \vec{\mu} \times \vec{B}$. Knowing that the magnetic moment for an electron is opposite to the angular momentum (because the charge is negative), does it predict the correct direction for the torque?

Problem 2.6 Estimating the splitting in a Stern-Gerlach experiment. The classical equations of motion for the Stern-Gerlach experiment yields the following results for the force on the current loop: (i) the force in the y -direction is zero; (ii) the force in the z -direction is a constant, proportional to the projection of the effective needle on the z -axis; and (iii) the force in the x -direction is oscillatory, with an oscillation

frequency ω . Consider a typical experimental magnet, which is 10 cm long and has a magnitude of 4,000 Gauss, then show that $\omega \approx 10^{11}$ rad/s. An atom emitted from a hot oven source will be moving at about 1000 m/s. Determine how long it will take to pass through the magnet. During this time, the force in the x -direction oscillates about a million times. Explain why this means it is a good approximation to take the average motion in the x -direction to be zero. Determine the magnitude of the splitting between an atom with angular momentum $\pm 2\pi\hbar$ on a screen that is 1 m away, assuming the gradient is 2000 Gauss/cm. Explain why the beam needed to be approximately 0.1 mm or thinner for the experiment to work.

Problem 2.7 Can a Stern-Gerlach experiment be performed on electrons? The Stern-Gerlach experiment is typically performed with atoms, which are electrically neutral. Explain why it is essentially impossible to do this experiment with electrons as opposed to atoms. Note further that even with atoms, not all will show a sizable effect. It turns out that silver is one of the best atoms to work with, and it is just luck that this is the atom Stern had been working on when he designed this experiment.

Problem 2.8 Three analyzer-loops in a row. We examine a set of experiments with three perpendicular analyzer loops where we have closed a different set of gates (see Figs. 2.21–2.26). For each figure, determine the state of the atom as it exits the final analyzer loop and determine the percentage of atoms that exit the final analyzer loop relative to the number that entered the first analyzer loop.

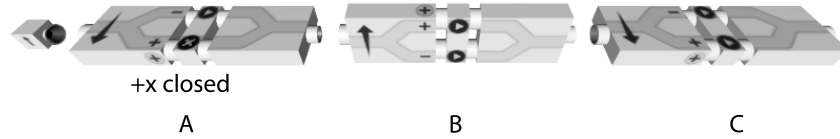


Fig. 2.21: Three analyzer experiment with the $+x$ gate on analyzer loop A closed.

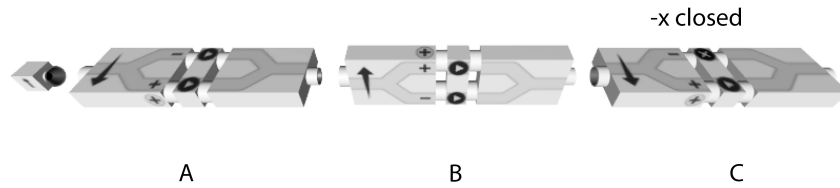


Fig. 2.22: Three analyzer experiment with the $-x$ gate on analyzer loop C closed.

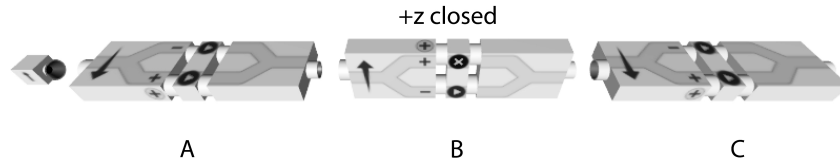


Fig. 2.23: Three analyzer experiment with the $+z$ gate on analyzer loop B closed.

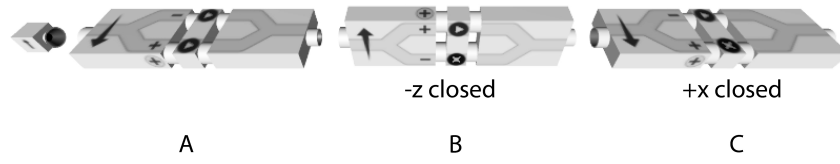


Fig. 2.24: Three analyzer experiment with the $-z$ gate on analyzer loop B closed and the $+x$ gate on analyzer loop C closed..

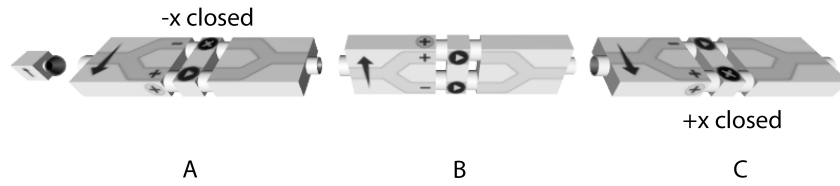


Fig. 2.25: Three analyzer experiment with the $-x$ gate on analyzer loop A closed and the $+x$ gate on analyzer loop C closed.

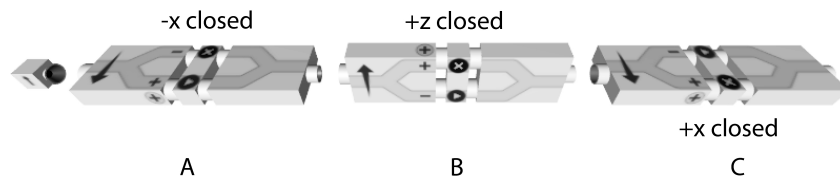


Fig. 2.26: Three analyzer experiment with the $-x$ gate on analyzer loop A closed, the $+z$ gate on analyzer loop B closed, and the $+x$ gate on analyzer loop C closed.

Problem 2.9 Quantum seeing in the dark. Something strange occurred when comparing the set-up in Fig. 2.25 with the set-up in Fig. 2.26. Adding a blockage in the middle analyzer loop in Fig. 2.26 increases the number atoms that exit! Write a paragraph explaining how such a result can occur in the quantum world. You may want to begin by explaining what happens in Fig. 2.25. It is the atoms that do not hit the new obstruction that tell us that the obstruction is there. This behavior is called “quantum seeing in the dark,” or an “interaction-free measurement.” One can detect the extra blockage with just a single atom if it hits none of the blockages. Be clear, concise, and persuasive in your paragraph.

Problem 2.10 The classical radius of the electron. A classical model for an electron is that it has a uniform charge density, with a total charge given by $-e$ and a radius r_c determined by the condition that we equate the electrostatic energy of a charged sphere to the rest-mass of the electron. The rest mass satisfies $mc^2 \approx 511,000$ eV. We compute the electrostatic energy as follows:

- (1) determine the radial electric field $\vec{E}$ of a uniform charged sphere of radius r . You should find Gauss’ law allows you to do this quickly.
- (2) The electrostatic energy is given by integrating the energy $\frac{1}{2}\epsilon_0|\vec{E}|^2$ over the sphere of radius r_c . Perform this integration to find the electrostatic energy as a function of e and r_c .
- (3) Equate the electrostatic energy to the rest mass and solve for an explicit formula for r_c in terms of constants, like e , m , and c .
- (4) Determine the quantitative value of the classical radius of the electron. You should find it is on the order of 1 fm.
- (5) Assume the sphere is spinning about its axis with an angular frequency of ω . Assume the mass density of the sphere is also uniform, so one can use the moment of inertia of the sphere to calculate the angular momentum. Equate this angular momentum to $\hbar$ to find the required angular velocity. How large is the linear velocity ωr_c at the equator. Compare this to the speed of light. (You should find it is significantly larger than the speed of light). You can ignore relativistic effects in your calculation.
- (6) Explain why the classical model of an electron as a uniform spinning sphere of charge is untenable.

Problem 2.11 Practice with the Dirac notation for vectors and inner products: Consider the following column vectors

$$|\psi\rangle \equiv \frac{1}{\sqrt{5}} \begin{pmatrix} -i \\ 2 \end{pmatrix} \quad \text{and} \quad |\phi\rangle \equiv \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ i \end{pmatrix}. \quad (2.34)$$

here, the $\equiv$ sign indicates the equivalence between an abstract vector on the left-hand side and a concrete representation of the column vector on the right-hand side.

- (1) Compute the corresponding representations for the bras $\langle\psi|$ and $\langle\phi|$. These must be row vectors (and do not forget the complex conjugation).
- (2) Verify that these two state vectors are normalized.

(3) Compute the inner product $\langle\psi|\phi\rangle$ and $\langle\phi|\psi\rangle$. Comment on the relationship between these two results.

Problem 2.12 Superposition. In the text, we described how a positive projection state on the z -axis could equivalently be written in terms of a superposition of states that have definite projections on the x axis via

$$|\uparrow\rangle_z = \frac{1}{\sqrt{2}} (|\uparrow\rangle_x + |\downarrow\rangle_x). \quad (2.35)$$

Using concrete experimental examples with analyzer-loops, explain in words what a superposition is.

Problem 2.13 Practice with Dirac notation and tensor products. We now examine inner products with tensor product states. Consider the state on the inside of the analyzer loop oriented in the x -direction, given by

$$\frac{1}{\sqrt{2}} |\uparrow\rangle_x \otimes |\text{front}\rangle + \frac{1}{\sqrt{2}} |\downarrow\rangle_x \otimes |\text{back}\rangle. \quad (2.36)$$

If we use the rules that $\langle\uparrow|\downarrow\rangle = 0$ for any direction of the magnetic field and $\langle\text{front}|\text{back}\rangle = 0$, and that each of these states is normalized, then we can compute the inner product of the state in Eq. (2.36) with the states $|\uparrow\rangle_z \otimes |\text{front}\rangle$ and $|\downarrow\rangle_z \otimes |\text{back}\rangle$. Also compute the overlaps with $|\downarrow\rangle_z \otimes |\text{front}\rangle$ and $|\uparrow\rangle_z \otimes |\text{front}\rangle$. Explain the general rule for when two tensor-product vectors must be orthogonal using the fact that the inner product of two tensor product vectors is the product of the inner products of its constituents.

Problem 2.14 Entanglement. Consider the entangled states we worked with for the Stern-Gerlach experiment. We said that they were always correlated, if the spin-state was $\uparrow$ then the spatial state is up for a vertical analyzer, while if the spin-state is $\downarrow$, then the spatial state is down for a vertical analyzer. Explain whether this means that if we measure it to be a positive projection on the z -axis, then do we know with no uncertainty what the spatial state is? What if we measure the spatial position. Do we know for sure what the spin state is? Explain in your own words what entanglement means to you in this context.

Problem 2.15 Requirements of quantum measurement. We stated earlier that a general provision of a quantum measurement is that it must have single quantum sensitivity and we needed to amplify the signal to be large enough to see it with classical measuring equipment. We discussed that the Stern-Gerlach experiment is typically not a single quantum experiment. What aspects of quantum measurement does it have nevertheless?

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